Math 15 — Unit 1: Assignments

The assignment from every lesson in the booklet. Printed work is not tracked on the website.

Lesson 1 Assignment *Show your work.*

Part A — Adding and Subtracting Integers

1. Fill in each blank, then evaluate.
(a) $-7+(-6)$: both numbers are negative (same sign), so add their values and keep the negative sign. $-7+(-6) =$ _____
(b) $10+(-3)$: the signs differ, so subtract the smaller size (distance from zero) from the larger and keep the sign of the number farther from zero. $10+(-3) =$ _____

2. Evaluate each expression.
(a) $-8+5$ (b) $6-11$ (c) $-9-(-2)$ (d) $-14+14$

3. Evaluate each expression.
(a) $12-20$ (b) $-15+(-8)$ (c) $7-(-9)$ (d) $-3-7$

4. A number starts at -5 on a number line. First it moves 8 units right, then 3 units left. Write an addition/subtraction expression for its final position, then evaluate it.

Part B — Multiplying and Dividing Integers

5. Fill in the blank, then evaluate. $(-6)(-4)$: both numbers are negative (same sign), so the product is _____ (positive/negative). $(-6)(-4) =$ _____

6. Evaluate each expression.
(a) $(-7)(5)$ (b) $(8)(-9)$ (c) $(-3)(-11)$ (d) $(6)(7)$

7. Evaluate each expression.
(a) $30 \div (-6)$ (b) $-45 \div (-9)$ (c) $-56 \div 8$ (d) $81 \div 9$

Part C — Order of Operations

8., 9., 10. Evaluate each expression using the correct order of operations.

8. (a) $5 + 3 \times (-4)$

8. (b) $-6 - 2 \times (-3)$

9. (a) $(2 - 9) \times (-2)$

9. (b) $-4 \times (3 + (-7))$

10. (a) $18 \div (-3) + 5$

10. (b) $-10 + 4 \times (-2) - (-3)$

Part D — Exponents with Integers

11. Evaluate each expression, and notice how the placement of the brackets changes the answer. **12.** Evaluate each expression using the correct order of operations.

11. (a) -5^2

11. (b) $(-5)^2$

12. (a) $3 + (-2)^2$

12. (b) $20 - 4 \times (-3)^2$

13. Evaluate each expression, showing every step in order.

(a) $-3 \times (4 - 7)^2 + 5$

(b) $(-18) \div (-2) - 3^2$

Part E — Applications

For 14 and 15, write an integer expression first, then evaluate it.

14. The temperature was $-6°$C at 7 a.m. By 1 p.m. it had risen $11°$C. By 9 p.m. it had dropped $9°$C from the 1 p.m. temperature. Write an integer expression for the 9 p.m. temperature, then evaluate it.

15. Malik's bank account balance was $-\$25$ (overdrawn by \$25). He deposited \$60, then withdrew \$18. Write an integer expression for his new balance, then evaluate it.

16. *(Multiple Choice)* Which of the following expressions is equal to 16? A. -4^2 B. $(-4)^2$ C. $-(4^2)$ D. $-2 \times (-4)^2$

Answer Key. 1. (a) -13 (b) 7. **2.** (a) -3 (b) -5 (c) -7 (d) 0. **3.** (a) -8 (b) -23 (c) 16 (d) -10. **4.** $-5 + 8 - 3 = 0$. **5.** positive; 24. **6.** (a) -35 (b) -72 (c) 33 (d) 42. **7.** (a) -5 (b) 5 (c) -7 (d) 9. **8.** (a) -7 (b) 0. **9.** (a) 14 (b) 16. **10.** (a) -1 (b) -15. **11.** (a) -25 (b) 25. **12.** (a) 7 (b) -16. **13.** (a) -22 (b) 0. **14.** $-6 + 11 - 9 = -4°$C. **15.** $-25 + 60 - 18 = \$17$. **16.** B.

Lesson 2 Assignment *Show your work.*

Part A — Equivalent Fractions and Simplifying

1. Fill in the missing numerator to make an equivalent fraction.
(a) $\frac{1}{2} = \frac{\ }{8}$ (b) $\frac{2}{3} = \frac{\ }{12}$ (c) $\frac{3}{5} = \frac{\ }{20}$

2. Simplify each fraction to lowest terms.
(a) $\frac{6}{8}$ (b) $\frac{9}{12}$ (c) $\frac{14}{21}$ (d) $\frac{20}{25}$

Part B — Operations with Fractions and Mixed Numbers

3. Add or subtract. Give your answer in lowest terms.
(a) $\frac{1}{3} + \frac{1}{6}$ (b) $\frac{3}{4} - \frac{1}{8}$ (c) $\frac{2}{5} + \frac{1}{2}$ (d) $\frac{5}{6} - \frac{1}{3}$

4. Add or subtract these mixed numbers.
(a) $1\frac{1}{2} + 2\frac{1}{4}$ (b) $3\frac{1}{3} - 1\frac{1}{6}$

5. Multiply. Simplify your answer.
(a) $\frac{1}{2} \times \frac{2}{5}$ (b) $\frac{3}{4} \times \frac{4}{9}$ (c) $\frac{5}{6} \times \frac{3}{10}$

6. Divide. Simplify your answer.
(a) $\frac{1}{3} \div \frac{2}{5}$ (b) $\frac{3}{4} \div \frac{1}{2}$ (c) $\frac{2}{5} \div \frac{4}{5}$

7. Multiply or divide these mixed numbers. (Convert to improper fractions first.)
(a) $1\frac{1}{2} \times \frac{2}{3}$ (b) $2\frac{1}{4} \div \frac{3}{4}$

Part C — Fractions and Decimals

8. Convert each fraction to a decimal.
(a) $\frac{1}{4}$ (b) $\frac{3}{5}$ (c) $\frac{7}{10}$ (d) $\frac{5}{8}$

9. Convert each decimal to a fraction in lowest terms.
(a) 0.5 (b) 0.25 (c) 0.35 (d) 0.4

10. For each fraction, state whether its decimal form *terminates* or *repeats*.
(a) $\frac{1}{4}$ (b) $\frac{4}{9}$ (c) $\frac{7}{8}$ (d) $\frac{2}{9}$

Part D — Order of Operations and Applications

11. Evaluate using the correct order of operations. Give your answer in lowest terms.
(a) $\frac{1}{2} + \frac{1}{3} \times \frac{3}{4}$ (b) $\frac{5}{6} - \frac{1}{4} \times \frac{2}{3}$

12. Evaluate using the correct order of operations.
(a) $3.2 - 0.5 \times 4$ (b) $(1.5 + 0.5) \times 2$

13. A pancake recipe calls for $\frac{3}{4}$ cup of milk to make one full batch. Priya wants to make only $\frac{1}{3}$ of the recipe to test it out. How much milk should she use?

14. Four friends go out for lunch. The total bill, including tax, is \$38.00. They split the bill evenly, and each friend also leaves a \$2.00 tip. How much does each friend pay in total?

15. *(Multiple Choice)* Which fraction below is equivalent to $\frac{3}{5}$?
A. $\frac{6}{15}$ B. $\frac{9}{15}$ C. $\frac{12}{15}$ D. $\frac{15}{15}$

16. *(Numerical Response)* Evaluate $0.6 \div 0.2 + 1.5$.
The answer is _____.

Answer Key. 1. (a) 4 (b) 8 (c) 12. **2.** (a) $\frac{3}{4}$ (b) $\frac{3}{4}$ (c) $\frac{2}{3}$ (d) $\frac{4}{5}$. **3.** (a) $\frac{1}{2}$ (b) $\frac{5}{8}$ (c) $\frac{9}{10}$ (d) $\frac{1}{2}$. **4.** (a) $3\frac{3}{4}$ (b) $2\frac{1}{6}$. **5.** (a) $\frac{1}{5}$ (b) $\frac{1}{3}$ (c) $\frac{1}{4}$. **6.** (a) $\frac{5}{6}$ (b) $1\frac{1}{2}$ (c) $\frac{1}{2}$. **7.** (a) 1 (b) 3. **8.** (a) 0.25 (b) 0.6 (c) 0.7 (d) 0.625. **9.** (a) $\frac{1}{2}$ (b) $\frac{1}{4}$ (c) $\frac{7}{20}$ (d) $\frac{2}{5}$. **10.** (a) terminates (b) repeats (c) terminates (d) repeats. **11.** (a) $\frac{3}{4}$ (b) $\frac{2}{3}$. **12.** (a) 1.2 (b) 4. **13.** $\frac{1}{4}$ cup. **14.** \$11.50. **15.** B. **16.** 4.5.

Lesson 3 Assignment *Show your work.*

Part A — Factors, Primes, and Prime Factorization

1. State all the factors of each number.
(a) 18 (b) 40

2. Classify each number as prime or composite.
(a) 31 (b) 42 (c) 53 (d) 57

3. Use a **division ladder** to determine the prime factorization.
(a) 92 (b) 88

4. Use a **factor tree** to determine the prime factorization.
(a) 78 (b) 99

Part B — GCF and LCM

5. Determine the greatest common factor by **listing factors**.
(a) 16 and 20 (b) 12, 18, and 30

6. Use **prime factorization** to determine the greatest common factor.
(a) 63 and 105 (b) 88 and 110 (c) 34, 51, and 85

7. Determine the lowest common multiple by **listing multiples**.
(a) 6 and 10 (b) 4, 5, and 8

8. Use **prime factorization** to determine the lowest common multiple.
(a) 16 and 20 (b) 21 and 35 (c) 9, 15, and 25

9. Determine both the greatest common factor *and* the lowest common multiple of 28 and 42.

Part C — Applications and Review

10. A coach has 40 water bottles and 32 granola bars to divide evenly among identical team bags, using every item, with the greatest number of bags possible. (a) How many team bags can she make? (b) How many water bottles and how many granola bars go in each bag?

11. Two rotating spotlights at a school dance sweep past the DJ booth on a regular schedule: Light A every 15 seconds, Light B every 20 seconds. If both lights just swept past the booth together, how many seconds will it be until they sweep past together again?

12. *(Multiple Choice)* The greatest common factor of 54 and 90 is
A. 6 B. 9 C. 18 D. 27

13. *(Numerical Response)* The lowest common multiple of 12, 16, and 24 is _____.

14. **Explain** why the greatest common factor of two *different* prime numbers (for example, 11 and 13) must always be 1, without doing any calculations.

Answer Key. 1. (a) $1, 2, 3, 6, 9, 18$ (b) $1, 2, 4, 5, 8, 10, 20, 40$. **2.** (a) prime (b) composite (c) prime (d) composite. **3.** (a) $92 = 2^2 \cdot 23$ (b) $88 = 2^3 \cdot 11$. **4.** (a) $78 = 2 \cdot 3 \cdot 13$ (b) $99 = 3^2 \cdot 11$. **5.** (a) 4 (b) 6. **6.** (a) 21 (b) 22 (c) 17. **7.** (a) 30 (b) 40. **8.** (a) 80 (b) 105 (c) 225. **9.** GCF = 14, LCM = 84. **10.** (a) 8 bags (b) 5 water bottles and 4 granola bars each. **11.** 60 seconds. **12.** C. **13.** 48. **14.** A prime's only factors are 1 and itself; two *different* primes share no factor but 1, so their GCF is 1.

Lesson 4 Assignment *Show your work.*

Part A — Naming the Sets

1. Given the set $\{-9,\ 0,\ 4,\ 17,\ -22,\ 33\}$, list the members that are:
(a) natural numbers (b) whole numbers (c) integers

2. Write each as a decimal, state whether the decimal terminates or repeats, and state whether the number is rational.
(a) $\frac{5}{8}$ (b) $\frac{2}{3}$ (c) -9 (d) 0.45 (as a fraction in lowest terms)

3. Decide whether each statement is true (T) or false (F).
(a) ___ Every integer is a rational number. (b) ___ Every rational number is an integer.
(c) ___ A terminating decimal cannot be written as a fraction. (d) ___ 0 is a whole number but not a natural number.
(e) ___ A repeating decimal is irrational. (f) ___ Every whole number is also an integer.

4. Classify each number as rational or irrational, and explain your reasoning.
(a) $\sqrt{6}$ (b) $\sqrt{16}$ (c) 2π (d) $0.808008000800008\ldots$

5. Complete the table. Place a checkmark in every set that each number belongs to.

Number	N	W	I	Q	$\overline{Q}$	R	**Number**	N	W	I	Q	$\overline{Q}$	R
21							0						
-5							1.8						
$\frac{3}{4}$							$-\sqrt{25}$						
$\sqrt{13}$							$0.\overline{72}$						

Part B — Reasoning About the Sets

6. Find one number that satisfies each condition.
(a) an integer, but not a whole number (b) a rational number, but not an integer (c) a real number, but not a rational number (d) a whole number, but not a natural number

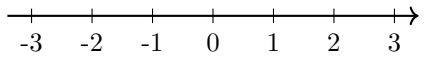
7. Complete each statement with *always*, *sometimes*, or *never*.
(a) A natural number is _____ a whole number. (b) A rational number is _____ an integer.
(c) The sum of two integers is _____ an integer. (d) The square root of a whole number is _____ irrational.
(e) A whole number is _____ a natural number.

8. Decide whether each statement is true (T) or false (F).
(a) ___ The set of natural numbers is nested inside the set of whole numbers. (b) ___ The set of rational numbers is nested inside the set of integers.
(c) ___ The set of integers is nested inside the set of rational numbers. (d) ___ The irrational numbers are nested inside the rational numbers.

Part C — Ordering, Explaining, and Multiple Choice

9. Order these numbers from least to greatest, then plot each one on the number line below.

$$-2.5, \quad \sqrt{5}, \quad 0, \quad -\sqrt{9}, \quad 1.5, \quad 0.3$$



10. In your own words, explain the difference between a rational number and an irrational number. Give one example of each.

11. *(MC)* Which of the following is an irrational number?

A. $\sqrt{25}$ B. $0.\overline{7}$
C. $\sqrt{10}$ D. $\dfrac{3}{4}$

12. *(MC)* The decimal representation of $\dfrac{7}{12}$ is

A. terminating and repeating
B. terminating and non-repeating
C. non-terminating and repeating
D. non-terminating and non-repeating

13. *(MC)* Which of the following is *not* rational?

A. $\sqrt{36}$ B. $\sqrt{40}$
C. -6 D. 0.65

14. Complete a diagram of the real number system like the one in the lesson: inside the rectangle for R, draw nested rectangles $N \subset W \subset I \subset Q$, and a separate box for $\overline{Q}$. Label every set and write one example number in each.



Answer Key. 1. (a) $4, 17, 33$ (b) $0, 4, 17, 33$ (c) all six. **2.** (a) 0.625, terminates, rational (b) $0.\overline{6}$, repeats, rational (c) -9.0, terminates, rational (d) $\frac{9}{20}$, terminates, rational. **3.** (a) T (b) F (c) F (d) T (e) F (f) T. **4.** (a) irrational (6 is not a perfect square) (b) rational ($\sqrt{16} = 4$) (c) irrational (d) irrational (pattern never repeats). **5.** 21: N, W, I, Q, R; -5: I, Q, R; $\frac{3}{4}$: Q, R; $\sqrt{13}$: $\overline{Q}, R$; 0: W, I, Q, R; 1.8: Q, R; $-\sqrt{25} = -5$: I, Q, R; $0.\overline{72}$: Q, R. **6.** (a) e.g. -4 (b) e.g. $\frac{1}{2}$ (c) e.g. $\sqrt{2}$ (d) 0. **7.** (a) always (b) sometimes (c) always (d) sometimes (e) sometimes. **8.** (a) T (b) F (c) T (d) F. **9.** $-\sqrt{9}\,(= -3)$, -2.5, 0, 0.3, 1.5, $\sqrt{5}\,(\approx 2.24)$. **10.** Rational: can be written $\frac{a}{b}$ with a, b integers, $b \neq 0$; its decimal terminates or repeats (e.g. $\frac{3}{4} = 0.75$). Irrational: decimal never ends and never repeats (e.g. $\sqrt{2}$). **11.** C. **12.** C. **13.** B. **14.** Rectangle R holding nested rectangles $Q \supset I \supset W \supset N$ and a separate box $\overline{Q}$ — as in the lesson diagram.

Lesson 5 Assignment *Show your work.*

Part A — Perfect Squares and Perfect Cubes

1. Evaluate each square root mentally.
(a) $\sqrt{9}$ (b) $\sqrt{36}$ (c) $\sqrt{100}$ (d) $\sqrt{4}$ (e) $\sqrt{169}$ (f) $\sqrt{64}$

2. Evaluate each cube root mentally.
(a) $\sqrt[3]{216}$ (b) $\sqrt[3]{343}$ (c) $\sqrt[3]{512}$ (d) $\sqrt[3]{729}$ (e) $\sqrt[3]{1000}$

3. State whether each number is a *perfect square*, a *perfect cube*, *both*, or *neither*.
(a) 49 (b) 64 (c) 90 (d) 1 (e) 200 (f) 125

Part B — Estimating Square Roots

4. State the two consecutive whole numbers each root lies between, and which one it is closer to.
(a) $\sqrt{30}$ (b) $\sqrt{70}$ (c) $\sqrt{110}$ (d) $\sqrt{58}$

5. Using the whole-number method (no calculator), estimate each square root to one decimal place.
(a) $\sqrt{18}$ (b) $\sqrt{85}$

6. Use a calculator to evaluate each square root to the nearest hundredth.
(a) $\sqrt{30}$ (b) $\sqrt{70}$ (c) $\sqrt{110}$

7. Use a calculator to evaluate each square root to the nearest tenth.
(a) $\sqrt{18}$ (b) $\sqrt{85}$

8. State whether each statement is true (T) or false (F).
(a) ____ $\sqrt{50}$ is between 7 and 8.
(b) ____ $\sqrt{50}$ is closer to 8 than to 7.
(c) ____ 100 is a perfect square.
(d) ____ 50 is a perfect square.

Part C — Applications and Review

9. A square rug covers an area of 45 ft^2. Estimate its side length to the nearest whole number, then use a calculator to find it to the nearest hundredth of a foot.

10. A cube-shaped water tank has a volume of 75 m^3. Estimate its edge length to the nearest whole number, then use a calculator to find it to the nearest hundredth of a metre.

11. *(Multiple Choice)* Which of the following is a perfect square?
A. 48 B. 81 C. 90 D. 150

12. *(Multiple Choice)* $\sqrt{68}$ lies between which two whole numbers?
A. 6 and 7 B. 7 and 8 C. 8 and 9 D. 5 and 6

13. *(Numerical Response)* To the nearest hundredth, $\sqrt{48} =$ _____.
(Record your answer from left to right.)

14. In your own words, explain how you decide which whole number a square root (of a non-perfect square) is closer to, once you know the two whole numbers it lies between.

Answer Key. 1. (a) 3 (b) 6 (c) 10 (d) 2 (e) 13 (f) 8. **2.** (a) 6 (b) 7 (c) 8 (d) 9 (e) 10. **3.** (a) perfect square (b) both (c) neither (d) both (e) neither (f) perfect cube. **4.** (a) between 5 and 6, closer to 5 (b) between 8 and 9, closer to 8 (c) between 10 and 11, closer to 10 (d) between 7 and 8, closer to 8. **5.** (a) ≈ 4.2 (b) ≈ 9.2. **6.** (a) 5.48 (b) 8.37 (c) 10.49. **7.** (a) 4.2 (b) 9.2. **8.** (a) T (b) F (c) T (d) F. **9.** ≈ 7 ft; $\sqrt{45} \approx 6.71$ ft. **10.** ≈ 4 m; $\sqrt[3]{75} \approx 4.22$ m. **11.** B. **12.** C. **13.** 6.93. **14.** Compare the distance from the number to each perfect square; the root is closer to the whole number whose perfect square is nearer (answers will vary).

Lesson 6 Assignment *Show your work.*

Part A — Vocabulary and Equivalence

1. State whether each radical is **entire** or **mixed**.
(a) $\sqrt{80}$ (b) $6\sqrt{5}$ (c) $\sqrt{54}$ (d) $2\sqrt{13}$

2. Evaluate $\sqrt{98}$ and $7\sqrt{2}$ as decimals (use a calculator), each rounded to three decimal places. What do you notice?

Part B — Entire Radicals to Mixed Radicals

3. Convert each entire radical to a mixed radical in simplest form.
(a) $\sqrt{12}$ (b) $\sqrt{28}$ (c) $\sqrt{99}$

4. Convert each entire radical to a mixed radical in simplest form.
(a) $\sqrt{80}$ (b) $\sqrt{147}$ (c) $\sqrt{175}$

5. Use prime factorization (a division ladder or a factor tree) to convert each entire radical to a mixed radical in simplest form.
(a) $\sqrt{132}$ (b) $\sqrt{156}$

Part C — Mixed Radicals to Entire Radicals

6. Convert each mixed radical to an entire radical.
(a) $3\sqrt{6}$ (b) $4\sqrt{7}$ (c) $6\sqrt{3}$

7. Convert each mixed radical to an entire radical.
(a) $8\sqrt{2}$ (b) $5\sqrt{5}$

Part D — Ordering, Applications, and Review

8. Without using a calculator, arrange the following radicals in order from least to greatest: $4\sqrt{3}$, $2\sqrt{10}$, $\sqrt{50}$, $3\sqrt{6}$.

9. Priya simplified $\sqrt{108}$ this way: $\sqrt{108} = \sqrt{4 \times 27} = 2\sqrt{27}$. Explain why this is not fully simplified, then finish simplifying it correctly.

10. Simplify $\sqrt{196}$. What is unusual about your answer, compared to the answers in Questions 3–5?

11. A square rug has an area of 176 square units. Determine the exact side length of the rug, in simplest mixed radical form.

12. *(Multiple Choice)* The correct simplified mixed radical form of $\sqrt{75}$ is
A. $5\sqrt{3}$ B. $3\sqrt{5}$ C. $25\sqrt{3}$ D. $5\sqrt{15}$

13. *(Numerical Response)* Convert $3\sqrt{10}$ to an entire radical: $\sqrt{}$.

14. **Explain** why $\sqrt{50}$ can be written as a mixed radical with a whole-number coefficient, but $\sqrt{51}$ cannot — without doing any long calculations.

Answer Key. 1. (a) entire (b) mixed (c) entire (d) mixed. **2.** Both ≈ 9.899 — the same value. **3.** (a) $2\sqrt{3}$ (b) $2\sqrt{7}$ (c) $3\sqrt{11}$. **4.** (a) $4\sqrt{5}$ (b) $7\sqrt{3}$ (c) $5\sqrt{7}$. **5.** (a) $2\sqrt{33}$ (b) $2\sqrt{39}$. **6.** (a) $\sqrt{54}$ (b) $\sqrt{112}$ (c) $\sqrt{108}$. **7.** (a) $\sqrt{128}$ (b) $\sqrt{125}$. **8.** $2\sqrt{10} < 4\sqrt{3} < \sqrt{50} < 3\sqrt{6}$. **9.** $27 = 9 \times 3$ still has a perfect-square factor, so $2\sqrt{27}$ is not finished; $\sqrt{108} = 6\sqrt{3}$. **10.** $\sqrt{196} = 14$ — a whole number, with no radical left. **11.** $4\sqrt{11}$ units. **12.** A. **13.** 90. **14.** $50 = 25 \times 2$ has the perfect-square factor 25; $51 = 3 \times 17$ has no perfect-square factor besides 1, so $\sqrt{51}$ is already in simplest form.

Lesson 7 Assignment *Show your work.*

Part A — Entire Cube Radicals to Mixed

1. State whether each radical is an entire radical or a mixed radical.
(a) $\sqrt[3]{35}$ (b) $6\sqrt[3]{2}$ (c) $\sqrt[3]{7}$ (d) $-4\sqrt[3]{10}$

2. Convert each radical to a mixed radical in simplest form.
(a) $\sqrt[3]{48}$ (b) $\sqrt[3]{32}$ (c) $\sqrt[3]{104}$ (d) $\sqrt[3]{56}$

3. Convert each radical to a mixed radical in simplest form.
(a) $\sqrt[3]{324}$ (b) $\sqrt[3]{81}$ (c) $\sqrt[3]{108}$ (d) $\sqrt[3]{297}$

4. Convert each radical to a mixed radical in simplest form.
(a) $\sqrt[3]{248}$ (b) $\sqrt[3]{448}$ (c) $\sqrt[3]{625}$ (d) $-\sqrt[3]{686}$

Part B — Mixed Cube Radicals to Entire

5. Convert each mixed radical to an entire radical.
(a) $2\sqrt[3]{7}$ (b) $3\sqrt[3]{4}$ (c) $4\sqrt[3]{6}$ (d) $5\sqrt[3]{6}$

6. Convert each mixed radical to an entire radical. *Watch the sign carefully.*
(a) $-2\sqrt[3]{5}$ (b) $-2\sqrt[3]{9}$ (c) $-3\sqrt[3]{3}$ (d) $-3\sqrt[3]{7}$

Part C — Fourth and Fifth Roots

7. Convert each radical to a mixed radical in simplest form.
(a) $\sqrt[4]{48}$ (b) $\sqrt[4]{80}$ (c) $\sqrt[5]{64}$ (d) $\sqrt[5]{160}$

8. Convert each mixed radical to an entire radical.
(a) $2\sqrt[4]{4}$ (b) $2\sqrt[4]{6}$ (c) $2\sqrt[5]{4}$ (d) $3\sqrt[5]{2}$

Part D — Both Directions and Review

9. Convert each radical to the requested form, in simplest form.

(a) $\sqrt[3]{152}$ (mixed) (b) $\sqrt[3]{405}$ (mixed) (c) $4\sqrt[3]{4}$ (entire) (d) $-4\sqrt[3]{9}$ (entire)

10. *(Multiple Choice)* $4\sqrt[3]{20}$ is equivalent to which entire radical?

A. $\sqrt[3]{1280}$ B. $\sqrt[3]{80}$ C. $\sqrt[3]{320}$ D. $\sqrt[3]{5120}$

11. *(Numerical Response)* The mixed radical $2\sqrt[3]{34}$ can be written as the entire radical $\sqrt[3]{n}$. The value of n is _____.

12. Explain why $\sqrt[4]{-16}$ is *not* a real number, but $\sqrt[3]{-16}$ can still be simplified. Then write $\sqrt[3]{-16}$ as a mixed radical in simplest form.

Answer Key. 1. (a) entire (b) mixed (c) entire (d) mixed. **2.** (a) $2\sqrt[3]{6}$ (b) $2\sqrt[3]{4}$ (c) $2\sqrt[3]{13}$ (d) $2\sqrt[3]{7}$. **3.** (a) $3\sqrt[3]{12}$ (b) $3\sqrt[3]{3}$ (c) $3\sqrt[3]{4}$ (d) $3\sqrt[3]{11}$. **4.** (a) $2\sqrt[3]{31}$ (b) $4\sqrt[3]{7}$ (c) $5\sqrt[3]{5}$ (d) $-7\sqrt[3]{2}$. **5.** (a) $\sqrt[3]{56}$ (b) $\sqrt[3]{108}$ (c) $\sqrt[3]{384}$ (d) $\sqrt[3]{750}$. **6.** (a) $\sqrt[3]{-40}$ (b) $\sqrt[3]{-72}$ (c) $\sqrt[3]{-81}$ (d) $\sqrt[3]{-189}$. **7.** (a) $2\sqrt[3]{3}$ (b) $2\sqrt[4]{5}$ (c) $2\sqrt[5]{2}$ (d) $2\sqrt[5]{5}$. **8.** (a) $\sqrt[4]{64}$ (b) $\sqrt[4]{96}$ (c) $\sqrt[5]{128}$ (d) $\sqrt[5]{486}$. **9.** (a) $2\sqrt[3]{19}$ (b) $3\sqrt[3]{15}$ (c) $\sqrt[3]{256}$ (d) $\sqrt[3]{-576}$. **10.** A. **11.** 272. **12.** An even index needs a non-negative radicand, since no real number to an even power is negative — so $\sqrt[4]{-16}$ is not a real number. An odd index allows a negative radicand: $\sqrt[3]{-16} = \sqrt[3]{-8 \times 2} = -2\sqrt[3]{2}$.

Lesson 8 Assignment *Show your work.*

Part A — Integer Operations and Order of Operations

1–2. Evaluate each expression.

1. (a) $-9 + 14$ (b) $-8 - 6$ (c) $5 - (-11)$ (d) $-20 + (-7)$

2. (a) $(-7)(4)$ (b) $(-45) \div (-9)$ (c) $6 \times (-8)$ (d) $-32 \div 4$

3. Evaluate each expression using the correct order of operations.

(a) $9 - 4 \times (-2)$ (b) $(-3 - 5) \times 2 + 7$ (c) -8^2 (d) $(-8)^2$

Part B — Fraction and Decimal Operations

4–5. Evaluate each expression. Try cancelling common factors first in 5(a).

4. (a) $\frac{1}{3} + \frac{3}{8}$ 4. (b) $\frac{7}{9} - \frac{1}{6}$ 5. (a) $\frac{5}{12} \times \frac{3}{10}$ 5. (b) $\frac{2}{5} \div \frac{3}{4}$

6. Convert and evaluate as directed.

(a) Convert $\frac{5}{8}$ to a decimal.

(b) Convert 0.24 to a fraction in lowest terms.

(c) Evaluate $\frac{3}{4} - \frac{1}{3} \times \frac{3}{5}$ in the correct order.

Part C — Prime Factorization, GCF, and LCM

7. Find the prime factorization of 126.

8. Use prime factorization to find the GCF of 54 and 72.

9. Use prime factorization to find the LCM of 6 and 16.

10. Bus Route A leaves the station every 15 minutes. Bus Route B leaves the station every 40 minutes. Both buses leave the station together at 8:00 a.m. How many minutes will it be before both buses leave the station together again? Give your answer in minutes, and then in hours and minutes.

Part D — Classifying and Ordering Real Numbers

11. For each number below, list every set it belongs to: natural (N), whole (W), integer (I), rational (Q), irrational ($\overline{Q}$), or real (R).

$-14, \quad \frac{7}{11}, \quad \sqrt{45}, \quad 49, \quad -\sqrt{25}, \quad 0.4\overline{18}$

12. Write the following numbers in order from least to greatest.

$-2.5, \quad \sqrt{3}, \quad -\sqrt{9}, \quad \frac{1}{2}, \quad 2$

Part E — Perfect Squares, Perfect Cubes, and Estimating Roots

13. Evaluate each root mentally.
(a) $\sqrt{121}$ (b) $\sqrt[3]{343}$

14. Estimate $\sqrt{55}$: state the two consecutive whole numbers it lies between, state which one it is closer to, and give a one-decimal estimate.

15. A square patio has an area of 130 ft^2. Estimate the side length to the nearest whole number, then use a calculator to find it to the nearest hundredth of a foot.

Part F — Simplifying Square-Root Radicals

16. Convert $\sqrt{98}$ to a mixed radical in simplest form.

17. Use prime factorization to convert $\sqrt{252}$ to a mixed radical in simplest form.

18. Convert $5\sqrt{6}$ to an entire radical.

19. Without using a calculator, arrange the following radicals in order from least to greatest: $4\sqrt{3}$, $2\sqrt{10}$, $\sqrt{50}$, $3\sqrt{5}$.

Part G — Simplifying Higher-Index Radicals

20. State whether each radical is entire or mixed.
(a) $\sqrt[3]{12}$ (b) $3\sqrt[3]{5}$

21. Convert $\sqrt[3]{250}$ to a mixed radical in simplest form.

22. Convert $2\sqrt[3]{7}$ to an entire radical.

23. Convert $\sqrt[4]{48}$ to a mixed radical in simplest form.

Part H — Mixed Challenge

24. Simplify $\sqrt{288}$ to a mixed radical. Then state whether the simplified value is rational or irrational, and explain how you know.

25. A community garden measuring 84 m by 126 m will be divided into the largest possible identical square plots, with no leftover strip along either side.
(a) Use the GCF of 84 and 126 to find the side length of each square plot.
(b) How many square plots will there be in total?

Answer Key. 1. (a) 5 (b) -14 (c) 16 (d) -27. **2.** (a) -28 (b) 5 (c) -48 (d) -8. **3.** (a) 17 (b) -9 (c) -64 (d) 64. **4.** (a) $\frac{17}{24}$ (b) $\frac{11}{18}$. **5.** (a) $\frac{1}{8}$ (b) $\frac{8}{15}$. **6.** (a) 0.625 (b) $\frac{6}{25}$ (c) $\frac{11}{20}$. **7.** $126 = 2 \times 3^2 \times 7$. **8.** 18. **9.** 48. **10.** 120 minutes = 2 hours 0 minutes. **11.** -14: I, Q, R; $\frac{7}{11}$: Q, R; $\sqrt{45}$: Q, R; 49: N, W, I, Q, R; $-\sqrt{25}$: I, Q, R; $0.4\overline{18}$: Q, R. **12.** $-\sqrt{9} < -2.5 < \frac{1}{2} < \sqrt{3} < 2$. **13.** (a) 11 (b) 7. **14.** $7 < \sqrt{55} < 8$; closer to 7; ≈ 7.4. **15.** Estimate ≈ 11; calculator ≈ 11.40 ft. **16.** $7\sqrt{2}$. **17.** $6\sqrt{7}$. **18.** $\sqrt{150}$. **19.** $2\sqrt{10} < 3\sqrt{5} < 4\sqrt{3} < \sqrt{50}$. **20.** (a) entire (b) mixed. **21.** $5\sqrt[3]{2}$. **22.** $\sqrt[3]{56}$. **23.** $2\sqrt[4]{3}$. **24.** $12\sqrt{2}$; irrational, since 2 is not a perfect square. **25.** (a) 42 m (b) 6 plots.